EEE4117F: AC Drives Laboratory Report

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II. Experimental Tasks

2.1 Experimental Data Collection

The table below presents the measured data for the induction motor operating at four supply frequencies (30, 40, 50, and 60 Hz) under five load torque settings (0%, 20%, 40%, 60%, 70% of rated torque).

Load Torque [%]	Active Power [W]3Ph	Reactive Power [VAR]3Ph	Apparent Power [VA]3Ph	cosϕ 3Ph	Voltage [V]	Current [A]	Speed [RPM]	Torque [Nm]
f = 30 Hz								
0	61.3	258.1	263.3	0.196	153.53	0.6	898	0.05
20	117.3	253.5	279.6	0.421	153.7	0.6	871	0.76
40	190.5	266.4	327.9	0.582	152.97	0.7	846	1.44
60	271.5	296.4	402.6	0.676	152.40	0.9	813	2.12
70	318.9	317.7	451.0	0.709	152.27	1.6	791	2.45
f = 40 Hz								
0	65.6	319.8	326.6	0.201	191.27	0.6	1192	0.07
20	153.9	314.8	350.6	0.439	190.67	0.6	1170	0.77
40	209.3	328.0	412.3	0.605	190.67	0.7	1146	1.45
60	352.7	359.5	503.9	0.700	190.27	0.9	1115	2.13
70	409.6	378.4	556.8	0.734	189.70	1	1097	2.45
f = 50 Hz								
0	75.1	369.9	377.8	0.199	222.03	0.6	1491	0.07
20	186.3	365.5	410.6	0.454	221.40	0.6	1469	0.77
40	303.2	376.4	483.6	0.627	221.53	0.7	1445	1.46
60	427.6	402	587.3	0.728	221.03	0.9	1416	2.14
70	493.7	419.5	648.2	0.761	221.17	1	1399	2.46
f = 60 Hz								
0	76.3	357.3	365.6	0.209	239.4	0.5	1790	0.09
20	210.8	348.6	407.6	0.517	231.07	0.6	1762	0.79
40	351.6	371.4	511.6	0.687	236.03	0.7	1730	1.47
60	506.9	409.6	652.0	0.777	233.40	0.9	1689	2.14
70	589.3	444.4	738.4	0.798	235.87	1	1667	2.47

Note: Power factor values recorded during the experiment are listed in the PF 3Ph column above. Computed power factor values are derived in Section 2.3.6.

2.2 Motor Characteristics

2.2.1 Torque/Speed Curves

The Torque–Speed curves for each supply frequency (30, 40, 50, 60 Hz) are shown in Figure 1 below. The corresponding Voltage-Speed plot is superimposed, with a trendline drawn through the no-load point of the Voltage-Speed curve at each frequency. For a base speed occurring at supply frequency of 50Hz, the constant-torque and constant-speed regions of operation are highlighted.

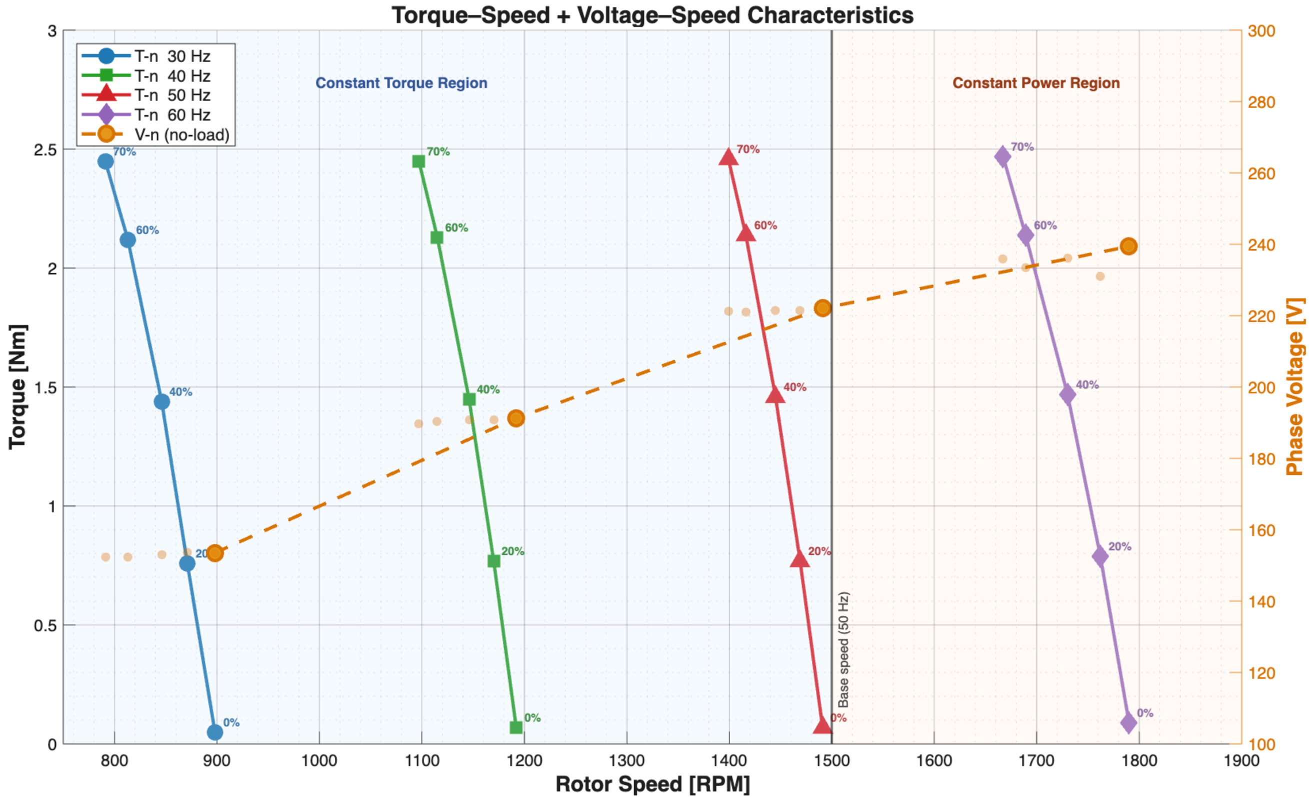


Figure 1: Torque/Speed and Voltage/Speed Plots

2.2.2 Voltage/Speed Curve

The Voltage–Speed characteristic is plotted on the same chart in Figure 1 above. The table below summarises the no-load (0%) voltage and speed for each frequency:

Frequency	Voltage [V]	Speed [RPM]	V/f Ratio [V/Hz]
30 Hz	153.53 V	898 RPM	5.118
40 Hz	191.27 V	1192 RPM	4.782
50 Hz	222.03 V	1491 RPM	4.441
60 Hz	239.4 V	1790 RPM	3.990

The near-constant V/f ratio confirms that the variable frequency drive maintains a constant volts-per-hertz ratio, which is characteristic of V/f (scalar) control. This keeps the air-gap flux approximately constant across frequencies.

2.3 Theory Related Questions

2.3.1 Torque Curve at Rated Current I = 1 A

At rated current I = 1 A, the electromagnetic torque of the induction motor can be estimated from the measured data points where the current equals approximately 1 A. From the data table, these operating points are:

Frequency [Hz]	Load [%]	Current [A]	Speed [RPM]	Torque [Nm]	Active Power [W]
30	70	1.6*	791	2.45	318.9
40	70	1	1097	2.45	409.6
50	70	1	1399	2.46	493.7
60	70	1	1667	2.47	589.3

** At 30 Hz and 70% load, current slightly exceeds 1 A (1.6 A). The nearest rated-current operating point at 30 Hz is at 60% load (I = 0.9 A).*

The results are superimposed on the Voltage-Speed curve and plotted on the figure below:

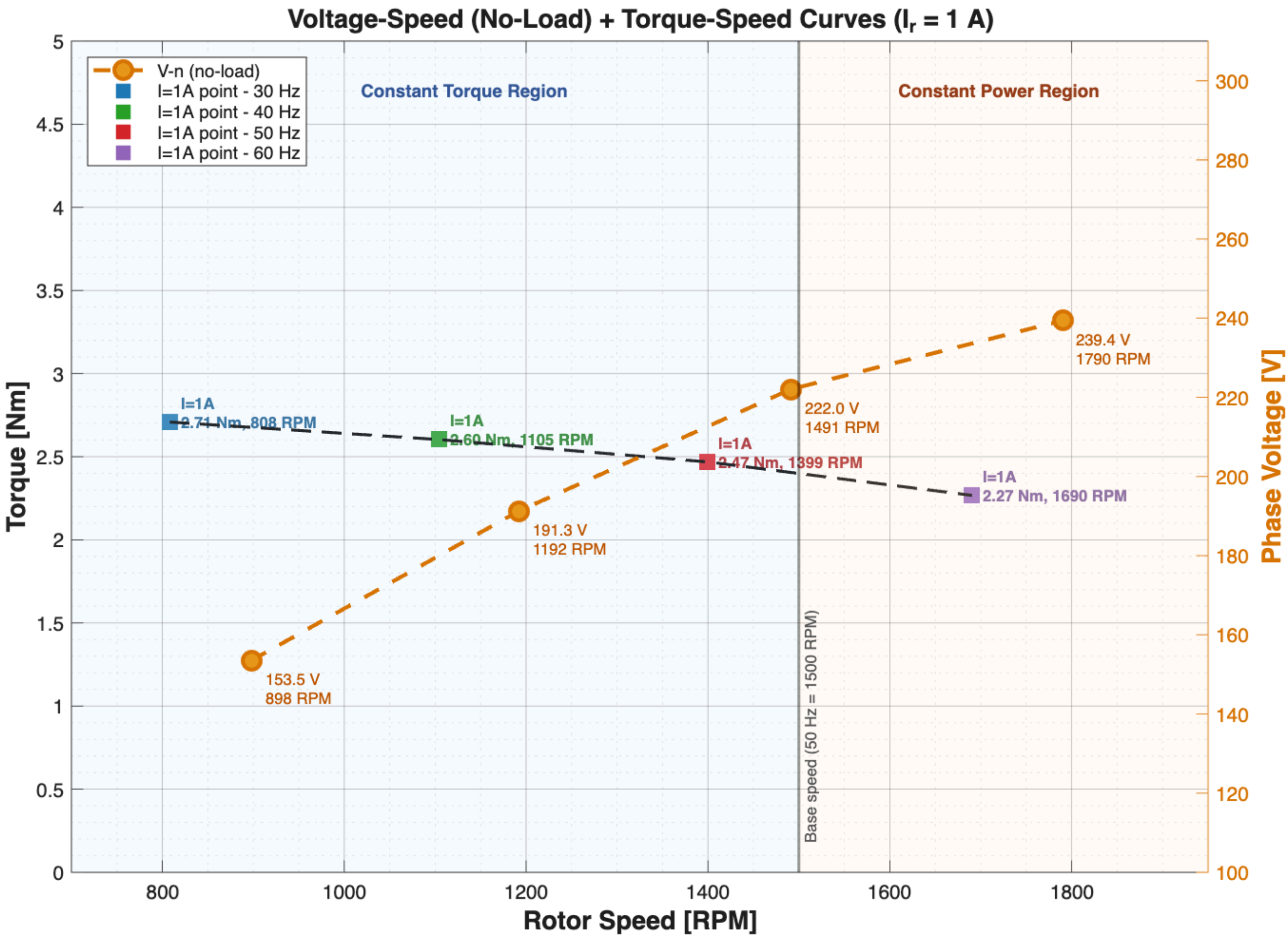


Figure 2: Voltage/Speed and Torque/Speed plots Superimposed

Comment: At the rated current, the torque remains approximately constant before the base speed. Which confirms that V/f control maintains constant flux in this region. Beyond the base speed, the torque drops due to field weakening. The motor cannot sustain the torque under the conditions beyond this point.

2.3.2 Constant Torque and Constant Power Regions

The constant-torque and power regions are shown on both figures above.

Constant Torque Region (below base speed):

In this region, the voltage and frequency increase proportionally, hence

$$\frac{V}{f} \approx \text{constant} \rightarrow \phi \approx \text{constant}$$

The electromagnetic torque of an induction motor is given by:

$$T = \frac{3}{\omega_s} \cdot \frac{(V^2/f)(R_2/s)}{(R_1+R_2/s)^2+(X_1+X_2)^2}$$

At low slip (near synchronous speed), the denominator is dominated by $(R_2/s)^2$, so:

$$T \approx \frac{3V^2R_2}{s \cdot \omega_s \cdot (R_2/s)^2}$$

Since V/f = constant, $V = k \cdot f$ and $\omega_s = 2\pi f$:

$$T \approx \frac{3k^2 f \cdot s}{2\pi R_2}$$

In the low-slip region, the air-gap flux held constant, the maximum torque is approximately constant. Hence below base speed (constant torque) the torque appears not to change as voltage increases.

2.3.3 Slip Speed η_{sl} at 30 Hz and 50 Hz

The per-unit slip is defined as:

$$\eta_{sl} = \frac{n_s - n}{n_s}$$

where $n_s = 60 \cdot \frac{f}{p}$ is the synchronous speed. For this motor (4-pole, p = 2):

n_s(30 Hz) = 60 × 30 / 2 = 900 RPM

n_s(50 Hz) = 60 × 50 / 2 = 1500 RPM

At rated current (I ≈ 1 A), the operating points from the superimposed curve are approximately 70% load at 50 Hz (1399 RPM) and 60% load at 30 Hz (813 RPM):

Frequency	ns [RPM]	n [RPM] at I≈1A	η_{sl}
30 Hz	900	813 (60% load)	0.0967
50 Hz	1500	1399 (70% load)	0.0673

Comment: The slip at 30 Hz is significantly higher than at 50 Hz. This is because the rotor resistance has a larger effect relative to the lower reactance, resulting in a larger slip for the same load torque. V/f drives are less efficient at lower frequencies.

2.3.4 Behaviour of ηsl in the Constant Power Region [1 mark]

η_{sl} increases in the constant power region. Above the base speed, the voltage is capped and the flux is weakened in this region so more slip is required to sustain torque.

2.3.5 How Braking is Achieved [1 mark]

In a VFD, regenerative braking can be achieved. The VFD reduces the output frequency below the rotor frequency. As a result, the synchronous speed drops below the rotor speed then the machine operates in the generator mode. Kinetic energy of the load is converted to electrical energy.

2.3.6 Power Factor: Computed vs Measured [10 marks]

The power factor is computed using the measured active and reactive power only (NOT the apparent power directly):

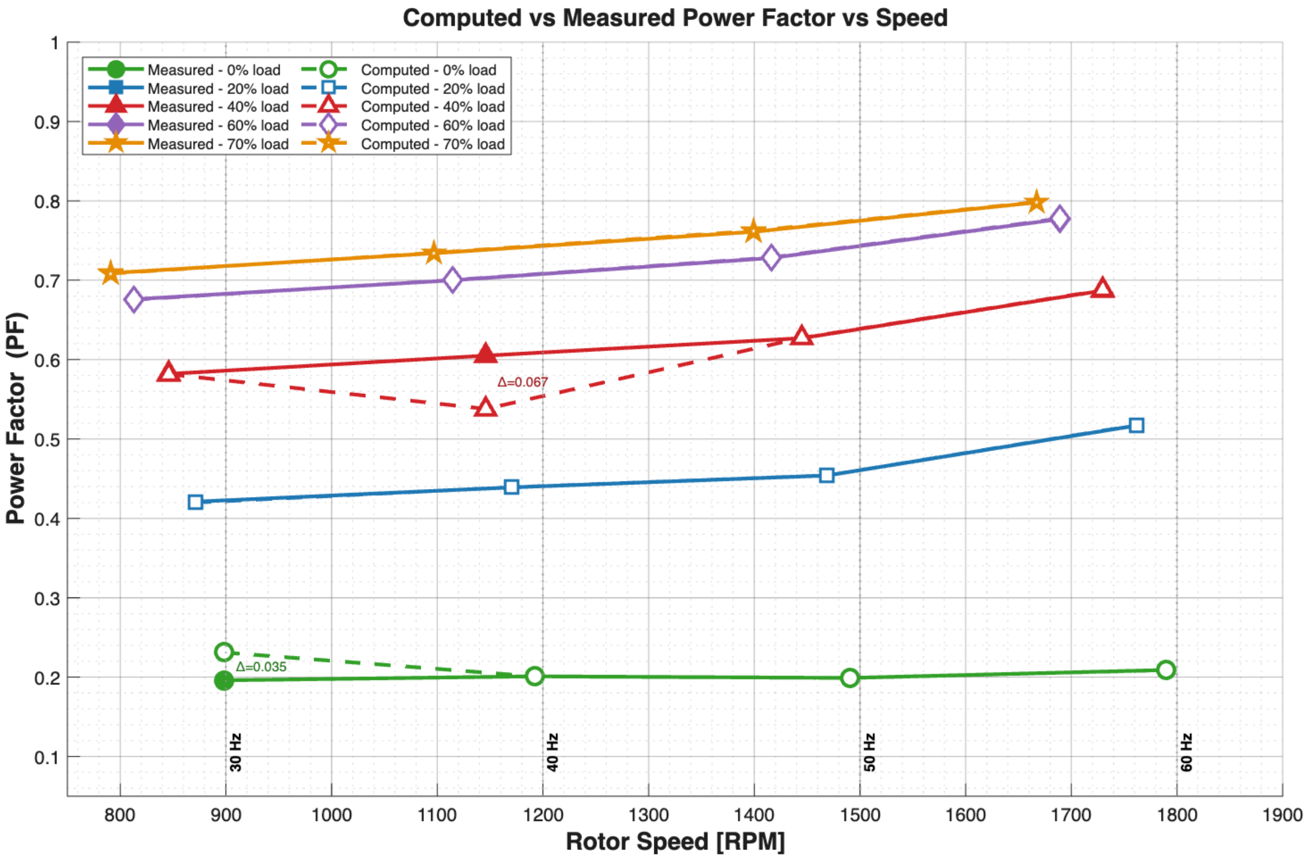
$$PF_{computed} = \frac{P}{\sqrt{P^2+Q^2}}$$

The table below presents both the measured and computed power factor for all operating points:

Freq [Hz]	Load [%]	Speed [RPM]	P [W]	Q [VAR]	PF Measured	PF Computed	|Difference|
30	0	898	61.3	258.1	0.196	0.231	0.035
30	20	871	117.3	253.5	0.421	0.420	0.001
30	40	846	190.5	266.4	0.582	0.582	0.000
30	60	813	271.5	296.4	0.676	0.675	0.001
30	70	791	318.9	317.7	0.709	0.708	0.001
40	0	1192	65.6	319.8	0.201	0.201	0.000
40	20	1170	153.9	314.8	0.439	0.439	0.000
40	40	1146	209.3	328.0	0.605	0.538	0.067
40	60	1115	352.7	359.5	0.700	0.700	0.000
40	70	1097	409.6	378.4	0.734	0.735	0.001
50	0	1491	75.1	369.9	0.199	0.199	0.000
50	20	1469	186.3	365.5	0.454	0.454	0.000
50	40	1445	303.2	376.4	0.627	0.627	0.000
50	60	1416	427.6	402.0	0.728	0.729	0.001
50	70	1399	493.7	419.5	0.761	0.762	0.001
60	0	1790	76.3	357.3	0.209	0.209	0.000
60	20	1762	210.8	348.6	0.517	0.517	0.000
60	40	1730	351.6	371.4	0.687	0.687	0.000

Freq [Hz]	Load [%]	Speed [RPM]	P [W]	Q [VAR]	PF Measured	PF Computed	|Difference|
60	60	1689	506.9	409.6	0.777	0.778	0.001
60	70	1667	589.3	444.4	0.798	0.798	0.000

The data is plotted in the figure below:

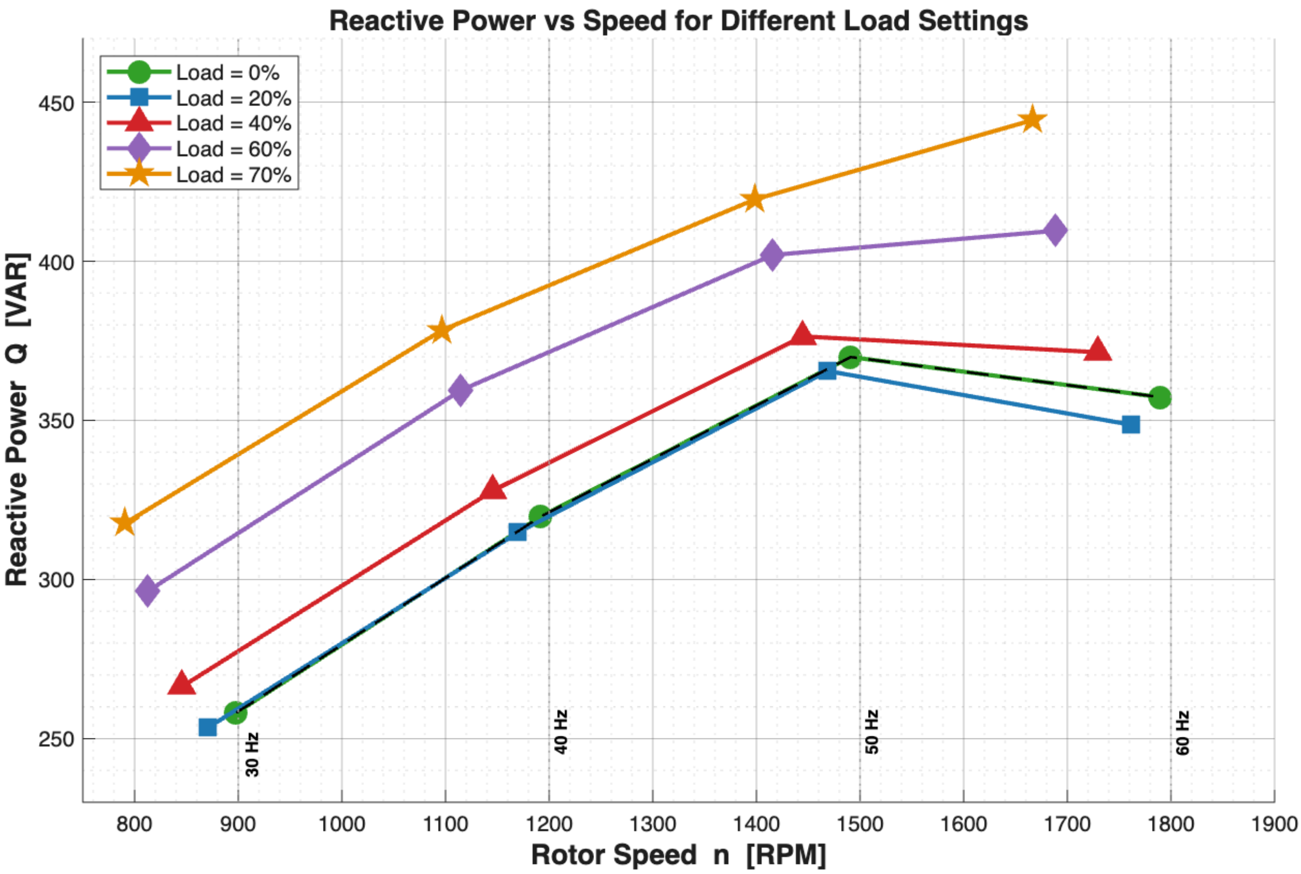


Comment on Results:

The above plot validates measurement accuracy. The computed and measured values match closely with a few small discrepancies due to rounding mismatch. Power factor increases with increasing load torque across all frequencies. The PF curves at different frequencies tend to converge at higher loads, which suggests that the reactive power stays constant and the active power scales with loading.

2.3.7 Reactive Power vs Speed for Different Load Settings

Freq [Hz]	Load [%]	Speed [RPM]	Q [VAR]
30	0	898	258.1
30	20	871	253.5
30	40	846	266.4
30	60	813	296.4
30	70	791	317.7
40	0	1192	319.8
40	20	1170	314.8
40	40	1146	328.0
40	60	1115	359.5
40	70	1097	378.4
50	0	1491	369.9
50	20	1469	365.5
50	40	1445	376.4
50	60	1416	402.0
50	70	1399	419.5
60	0	1790	357.3
60	20	1762	348.6
60	40	1730	371.4
60	60	1689	409.6
60	70	1667	444.4



Comment on Results:

As load increases, the reactive power increases slightly. This is due to the leakage reactances (stator and rotor) as they contribute more as the load increases. Reactive power is mainly from the magnetising reactance.